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LAMMPS-GUI: A Cross-Platform Graphical Tool to Learn and Explore Molecular Dynamics with LAMMPS

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Summary

LAMMPS is a widely used, massively parallel classical molecular dynamics (MD) engine for particle-based simulations at the atomic, mesoscopic, and continuum scales (Thompson et al., 2022). Implemented in C++ as a no-frills, console-mode application, it is highly portable and efficient, but learning to use it requires mastering separate – and often platform-specific – tools for editing inputs, plotting thermodynamic data, and visualizing atomic configurations *before* one can begin to learn MD itself.

LAMMPS-GUI is a cross-platform graphical application, written in C++17 using version 6 of the Qt framework (The Qt Company, 2026), that combines these tasks in one program: a syntax-highlighting input editor with auto-completion and documentation lookup, live execution with real-time output monitoring, interactive thermodynamic charts, a snapshot image viewer, and a slide-show viewer. Rather than launching LAMMPS as an external process, it embeds the engine through the C-language library interface (FrantzDale et al., 2010), running the simulation in a concurrent worker thread so the interface stays responsive and the run can be monitored, plotted, visualized, and cleanly stopped. By mirroring the traditional "edit, run, observe, analyze" workflow, it lets users move freely between the graphical tool and the command-line executable – essential once a workflow outgrows a personal computer. Pre-compiled packages for Windows, macOS, and Linux let most users start instantly, without compiling anything.

Statement of need

LAMMPS-GUI grew out of teaching LAMMPS at tutorials and workshops, where much of the available time went not to molecular dynamics but to installing and operating the surrounding toolchain – a text editor, a plotting program, and a molecular visualization package – most of which differ between Windows, macOS, and Linux. Earlier workarounds were either too complex to sustain or broke when Apple moved to the ARM architecture. As a single, self-contained, cross-platform tool that mirrors the console workflow, LAMMPS-GUI removes this barrier, letting instructors teach one interface on all platforms and focus on LAMMPS and molecular dynamics itself.

State of the field

Other approaches to making LAMMPS more accessible exist, but they target different needs. Commercial molecular-modeling packages provide structure editors that export LAMMPS inputs and re-import results; Atomify compiles a modified LAMMPS to WebAssembly and runs simulations and visualization inside a web browser (Hafreager, 2025); and dedicated visualization programs such as OVITO (Stukowski, 2010) and VMD (Humphrey et al., 1996)

render trajectories with a breadth and quality that LAMMPS-GUI does not attempt to match. None of these, however, integrates input editing, live simulation, output monitoring, plotting, and visualization into one application that follows the same edit, run, observe, analyze loop as the standalone executable. LAMMPS-GUI fills that gap. It also imports tutorial materials, such as the official LAMMPS tutorials (Gravelle et al., 2025) on molecular soft matter, with a collection on materials science available in beta and on discrete element modeling in preparation. While it is aimed primarily at beginners, features such as rapid prototyping of input decks, debugging failing inputs by visualizing selected components, and interactive construction of reproducible `dump image` commands also make it useful to experienced researchers.

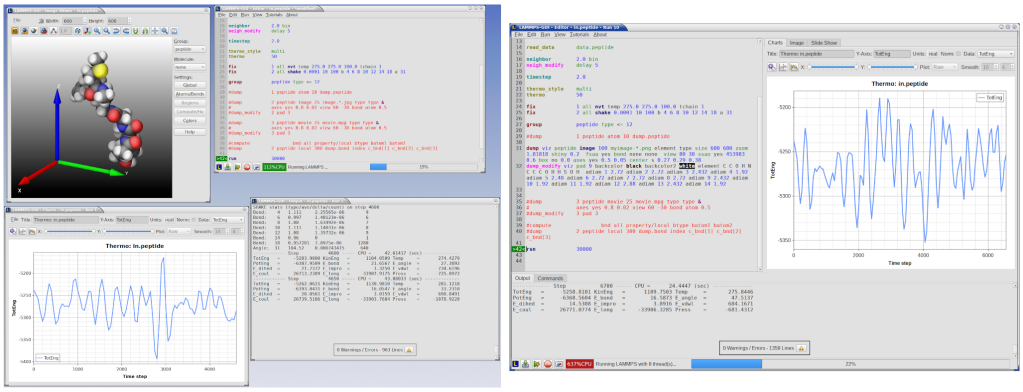


Figure 1: A LAMMPS-GUI session with a simulation in flight, in the individual window mode (left) and the combined window mode (right). The editor shows syntax highlighting, line numbers, and a marker on the current input line; the status bar shows CPU utilization and run progress; the Image Viewer shows the starting geometry, the Output window the screen output, and the Charts window a live plot of a thermodynamic column. In the combined window mode these views are tabs beside and below the editor.

Software design and functionality

LAMMPS-GUI follows an object-oriented design in which a central window coordinates largely self-contained components, all access to LAMMPS being funneled through a single adapter class. Two viewing modes are offered (see Figure 1): the original individual window mode, where every component is a window of its own, and a recently added combined window mode, where the views are docked as tabbed panels beside and below the editor in one main window, separated by movable splitters. Key capabilities of LAMMPS-GUI include:

- **Editing.** A LAMMPS-aware editor with syntax highlighting, per-command-category auto-completion, in-place documentation lookup, block comment/uncomment, and command reformatting.
- **Running.** Execution of the editor buffer through the library interface with no intermediate file I/O, a progress bar with estimated completion, support for the GPU, INTEL, KOKKOS, and OPENMP accelerator packages, and recovery from simulation errors – reported as catchable exceptions – without crashing.
- **Output and charts.** An output window that highlights warnings and errors and makes documentation URLs clickable. A charts window that plots thermodynamic data read directly from the running simulation (not scraped from text) or from external files, with Savitzky-Golay smoothing (Savitzky & Golay, 1964). Charts data post-processing such as autocorrelation functions, Fourier transforms, static structure factors, and polynomial, Birch-Murnaghan equation-of-state, and custom nonlinear fits.
- **Visualization.** A snapshot image viewer that builds high-quality renderings via the LAMMPS `dump image` command, with interactive controls and the ability to copy the generated command back into the script for reproducible figures, plus a slide-show viewer

that plays image sequences, imports frames from movie files, and exports movies.
- **Tutorials and convenience.** A guided wizard that downloads lesson inputs from several tutorial collections, a tabbed preferences dialog, hand-off of the current system to OVITO and VMD, and command-line flags exposing the charting, slide-show, and text viewers as standalone utilities.

What LAMMPS-GUI deliberately does not provide is a molecular editor or builder. Creating complex, physically sound initial configurations with their topology has never been within the scope of LAMMPS, and the GUI does not address it either. The built-in lattice, region, and molecule template commands cover many cases; beyond those, and especially for the atom typing and partial charge assignment that molecular force fields require, force-field-specific external tools are needed, for which the LAMMPS website maintains a curated list (The LAMMPS Developers, 2026a).

A recurring theme is the co-evolution of LAMMPS-GUI and LAMMPS: front-end needs drove improvements to the engine and its library interface. For example: catchable exceptions, a locked cache of live thermodynamic data, and greatly expanded snapshot rendering, collected into a dedicated GRAPHICS package (Kohlmeyer & Berger, 2025) to bring functionality from dedicated visualization tools directly to a running simulation without requiring a saved trajectory. In its default *plugin mode*, LAMMPS-GUI loads the shared library at run time with no link-time dependency, so one binary can pair with – or download – different LAMMPS builds.

Research impact statement

LAMMPS-GUI has shipped with LAMMPS since the August 2023 stable release, first from within the LAMMPS source tree and since August 2025 as an independently versioned package that the LAMMPS build system fetches on request; the pre-compiled LAMMPS packages for Windows, macOS, and Linux posted with each LAMMPS release bundle it (The LAMMPS Developers, 2026b). It is the primary tool of the peer-reviewed LAMMPS tutorials (Gravelle et al., 2025), which are written around it, and it has anchored public hands-on training since the 2024 LAMMPS master class, including the tutorial session of the 2025 LAMMPS Workshop and Symposium (The LAMMPS Developers, 2025); recordings of both live streams are online (The LAMMPS Developers, 2024, 2025). Its development has also fed improvements back into the LAMMPS engine itself, as described above. The project is maintained as research software with a documentation site (Kohlmeyer, 2026a), releases archived on Zenodo (Kohlmeyer, 2026b), continuous-integration builds on three platforms, an automated test suite, and user support through a dedicated tag on the LAMMPS forum (The LAMMPS Developers, 2026c).

AI usage disclosure

The design and architecture of LAMMPS-GUI were developed without the use of generative AI. However, AI-based coding assistants such as GitHub Copilot and Claude Code were used more recently to support the refactoring and modularization of the existing code base, and to assist with editing this manuscript. All AI-generated suggestions were reviewed, tested, and revised by the author, who takes full responsibility for the software and for the content of this paper.

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